

Question 1: Identify Data Types

Create variables for the following values:

- Your name
- Your age
- Your height
- Whether you are a student
- Your favorite programming language

Print each variable along with its data type.

Question 2: Type Conversion

Create the following variables:

```
num = 25
price = 49.95
word = "100"
flag = True
```

Convert:

- Integer to float
- Float to integer
- Integer to string
- String to integer
- Boolean to integer

Print each converted value and its data type.

Question 3: User Input Conversion

Accept the following inputs from the user:

- Name
- Age
- Salary

Convert the inputs into appropriate data types and display:

```
Name :
Age :
Salary :
```

Also display the data type of each variable.

Question 4: Mixed Data Types

Create a list containing the following values:

```
10
3.14
"Python"
True
```

Using indexing, print each element along with its data type.

Question 5: Tuple Practice

Create a tuple containing five numbers.

Display:

- First number
 - Last number
 - Third number
 - Total number of elements
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Question 6: Dictionary Practice

Create a dictionary containing:

- Name
- Age
- City

Print:

- Entire dictionary
- Name
- Age
- City

using their keys.

Question 7: Nested Data Types

Create the following dictionary:

```
student = {  
    "name": "Rahul",  
    "marks": [75, 82, 91],  
    "city": "Pune"  
}
```

Display:

- Student name
 - Second mark
 - City
-

Question 8: Boolean Expressions

Create variables:

```
a = 25  
b = 40
```

Print the result and data type of the following expressions:

- $a > b$
 - $a < b$
 - $a == b$
 - $a != b$
 - $a \geq 20$
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Question 9: Mutable vs Immutable

Create:

```
list1 = [10, 20, 30]  
tuple1 = (10, 20, 30)
```

Change the first element of the list.

Then try changing the first element of the tuple.

Observe and explain the output.

Question 10: Nested List

Create the following list:

```
data = [  
    [10,20],  
    [30,40],  
    [50,60]  
]
```

Print:

- 10
- 40
- 60

using indexing only.